

(G7) RECURSIVE LATENT ALIGNMENT: MULTI-SCALE CLIMATE DOWNSCALING VIA FOUNDATION EMBEDDINGS

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1 Abstract

We propose a framework for probabilistic precipitation downscaling from 25 km ERA5 data to 1.5 km resolution over Illinois, conditioned on AlphaEarth embeddings as terrain-aware covariates. Our approach recursively applies a learned 2x super-resolution operator built with a Conditional Flow Matching (CFM) model, generating large ensembles of downscaled data via noise injection. We establish baselines using bicubic interpolation and XGBoost (with and without AlphaEarth conditioning) and validate all models against 800 m PRISM observations.

2 Introduction and Motivation

High-resolution precipitation fields are critical for hydrological modeling [1], agricultural planning [2], and infrastructure risk assessment [3]. Global reanalysis products such as ERA5 [4] provide physically consistent atmospheric states at 25 km resolution, but are far too coarse for applications that require sub-kilometer detail. Existing work in dynamical downscaling via regional climate models addresses this gap, but is prohibitively expensive for ensemble generation.

This project investigates the potential benefits of AlphaEarth (AEF) [5] embeddings in a deep learning-based probabilistic downscaling framework. AEF provides dense, multi-modal representations of the Earth’s surface encoding topography, land cover, soil type, hydrology, and vegetation phenology, offering a richer terrain context than the elevation-only conditioning used in prior work [6]. Since this project explores a potential downstream task of GeoFM-generated embeddings and designs a flow matching-based deep-learning model [7], we believe it aligns well with the goals of this course.

3 Related Work

Traditional dynamical downscaling employs regional climate models (RCMs), which are computationally intensive and typically deterministic [8]. Machine learning approaches have increasingly replaced RCMs for statistical downscaling, but most focus on deterministic mappings (CNNs, U-Nets) that cannot provide uncertainty quantification [9]. Recent probabilistic approaches use diffusion models [10, 11] and flow matching [12] to generate ensembles, but are trained on RCM-generated data and output at kilometer scales. GenBCSR [12] and R2-D2 [6] both adopt residual framing - modeling the difference between bicubic-upsampled coarse fields and the fine target - which reduces the dynamic range of the generative target. R2-D2 conditions on terrain height and sub-grid orographic variance; our approach generalizes this by using the full AEF embedding vector as terrain conditioning. Geospatial Foundation Models (GeoFMs) like Prithvi [13], Clay [14], and AlphaEarth have demonstrated strong performance on downstream tasks; however, to our knowledge, no prior work has used GeoFM embeddings as conditioning for climate downscaling.

4 Methodology

4.1 Data Preprocessing Pipeline

We process three data sources for the Illinois domain (36.97°–42.51°N, 91.51°–87.50°W):

- **ERA5** (25 km, 0.25°): Hourly accumulated precipitation from two daily forecast initializations (00Z, 12Z), each with 12 accumulation steps. We sum the two 12-hour totals to obtain daily precipitation on a 23×17 grid. We also use **ERA5-Land** (9 km, 0.1°), containing precipitation on a 56×41 grid, regridded to a 0.125° grid (45×33) to serve as the 12.5 km fine-resolution target.

- **AlphaEarth Foundation Embeddings:** 64-dimensional embeddings at 10 m resolution, streamed from a >400 GB Zarr store on Source Cooperative [15]. We apply GeM pooling ($p = 3$) (recently explored as a high-performing yet computationally cheap pooling strategy [16]) to produce multi-resolution copies at 50, 25, 12.5, 6.25, 3.125, and 1.5625 km for each of 9 years (2017–2025).
- **PRISM** [17] (800 m, 0.008°): Daily accumulated precipitation regridded to a 1.5625 km grid (368×272) to serve as the fine-resolution validation.

We use this data to construct two resolution pairs. Both pairs use 2017–2024 for training (2,922 days) and 2025 for testing (366 days):

- **Pair A** (50 km $\rightarrow$ 25 km): The 25 km ERA5 field is coarsened by $2 \times$ average pooling to 50 km, then bicubic-upsampled back to the 25 km grid. The residual $r_A = \text{ERA5}_{25} - \text{BicubicInterp}(\text{ERA5}_{50})$ is the training target.
- **Pair B** (25 km $\rightarrow$ 12.5 km): ERA5-Land is regridded to 12.5 km. ERA5 at 25 km is bicubic-upsampled to the 12.5 km grid. The residual $r_B = \text{ERA5}_{\text{Land}}_{12.5} - \text{BicubicInterp}(\text{ERA5}_{25})$ is the training target.
- **Pair C** (3.125 km $\rightarrow$ 1.5625 km): The model’s recursively-generated output at 3.125 km is stored. The residual $r_C = \text{PRISM}_{1.5625} - \text{BicubicInterp}(\text{Model}_{3.125})$ is the training target.

4.2 Data Augmentation

Data augmentation was used to create additional training samples, since < 3000 data samples are far too few to train the CFM model (3M parameters). We adapt the augmentation technique in [18] by generating, from each of the top 40% of rainy days, additional data samples via horizontal and vertical flips, and their combination. In this way, half of the data becomes 4x the size.

4.3 Baseline Models

Given the large scale of AEF embeddings and the time spent applying for the DeltaAI compute allocation, we selected XGBoost as our baseline due to its computational efficiency and strong performance on high-dimensional tabular inputs. We evaluate three baselines:

- **Bicubic interpolation:** Predicts residual = 0 (i.e., fine field = coarse upsampled).
- **XGBoost without AEF:** Per-pixel regression with a single features: $\log(1 + p_{\text{coarse}})$.
- **XGBoost with AEF:** Adds 128 features (64 coarse-scale + 64 fine-scale AEF embedding dimensions) for a total of 129 features per pixel. Both XGBoost models train on log-space residuals with reduced complexity ($\text{max_depth} = 3$, $\text{min_child_weight} = 50$) to prevent overfitting on small residuals.

4.4 CFM Model Design

Our main model uses Conditional Flow Matching (CFM) to learn a conditional distribution over fine-scale residuals. The velocity field v_θ is parameterized by a U-Net with encoder channels [48, 96, 192], one residual block per level with GroupNorm, and multi-head attention (2 heads) at every level. The input concatenates $[x_{\text{coarse_up}}, x_t]$ along the channel dimension, where $x_t = (1 - t) \cdot z + t \cdot r_{\text{target}}$ is the flow interpolant between Gaussian noise $z \sim \mathcal{N}(0, I)$ and the target residual $r_{\text{target}} = x_{\text{fine}} - x_{\text{coarse_up}}$. Time t is encoded via sinusoidal embeddings followed by an MLP, and injected additively into each residual block.

Dual-resolution cross-attention. At each U-Net level, the hidden state h_i provides queries, while AEF embeddings at both the coarse and fine resolution provide keys and values:

$$h_{i+1} = h_i + \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V, \quad Q = W_q h_i, \quad K/V = W_{kv} [\alpha_{\text{coarse}}; \alpha_{\text{fine}}] \quad (1)$$

Unlike the XGBoost baseline where AEF enters as static per-pixel features, cross-attention allows the model to dynamically modulate AEF influence based on precipitation state and spatial context.

We use OT-CFM (via TorchCFM) [19] with $\sigma = 0$, since the optimal transport objective leads to faster and more accurate flows, allowing for efficient inference.

4.5 Training

Training Phase 1 (250 epochs) learns scale-invariant $2\times$ residual generation on pair A and B data (50→25 km and 25→12.5 km) with three losses: flow matching ($\mathcal{L}_{\text{FM}} = \|v_\theta(t, x_t) - u_t\|^2$, weight 1.0), cycle-consistency enforcing mass conservation ($\mathcal{L}_{\text{cycle}} = \|\text{AvgPool}(\hat{x}_{\text{fine}}) - x_{\text{coarse}}\|^2$, weight 0.8), and spectral PSD matching (weight 0.4, activated from epoch 5). AEF cross-attention weights are frozen for the first 10 epochs.

Training Phase 2 (50 epochs) builds rollout robustness by constructing 2-step sequences from paired Pair A and Pair B data (50→25→12.5 km). The model’s Pair A output is used as input for the Pair B step, and gradients flow through both steps. A rollout consistency loss is added, which pools the 12.5 km output back to 50 km and compares against the original ERA5 coarse field ($\mathcal{L}_{\text{rollout}} = \|\text{AvgPool}(\hat{x}_{12.5, 4\times}) - x_{50}\|^2$, weight 0.1). AEF projections remain frozen. LR is set to 10% of Phase 1’s final value.

Training Phase 3 (5 epochs) performs PRISM-supervised fine-tuning to correct biases in ERA5 data. The model runs a 4-step recursive rollout from 25 km to 1.5625 km. Steps 0–2 execute without gradients; only step 3 (3.125→1.5625 km) is differentiable. The loss is calculated as $\mathcal{L}_{\text{PRISM}} = \text{MSE}(\hat{x}, x_{\text{PRISM}}) + 0.5 \cdot \text{MSE}(\log(1 + \hat{x}), \log(1 + x_{\text{PRISM}}))$. To prevent collapse, each PRISM gradient step is followed by three flow-matching steps on Pair A/B data.

4.6 Recursive Ensemble Inference

At inference, starting from 25 km ERA5, we apply four recursive $2\times$ downscaling steps to reach 1.5625 km. At each step, the coarse field is bicubic-upsampled to $2\times$ resolution, then the CFM model integrates a noise sample $z^{(i)} \sim \mathcal{N}(0, I)$ from $t = 0$ to $t = 1$ via Euler integration to produce a predicted residual. Different initial noise vectors $z^{(i)}$ produce different but spatially coherent ensemble members, capturing the inherent ambiguity in fine-scale precipitation structure. We generate N -member ensembles at each step, with each member’s output serving as input to the next step. For the XGBoost baseline, ensembles are instead generated via empirical noise injection: $\hat{x}^{(i)} = \text{expm1}(\log(1 + x_{\text{coarse_up}}) + \hat{r} + \epsilon^{(i)})$ where $\hat{r}$ is the deterministic XGBoost residual and $\epsilon^{(i)}$ is sampled from the training residual distribution at each scale.

4.7 Validation

We validate against PRISM 2025 daily precipitation at ~ 800 m resolution. PRISM TIF files are cropped to Illinois and bilinearly regridded to the 1.5625 km output grid. To assess performance across precipitation regimes, we categorize each test day by rain presence in both data sources (domain-mean threshold of 0.1 mm/day): *both_rain* (ERA5 and PRISM rainy), *era5_only* (ERA5 rainy, PRISM dry), and *prism_only* (PRISM rainy, ERA5 dry). We report metrics both per-category and aggregated: 1) **RMSE** of the ensemble mean against PRISM, 2) **CRPS**, the traditional scoring rule for probabilistic forecasts [20, 21], 3) **Ensemble coverage** at the 50%, 80%, and 90% levels (% of PRISM pixels falling within the ensemble prediction interval), and 4) **Spread-skill ratio**, the mean ensemble standard deviation divided by mean absolute error.

5 Results

5.1 Baseline Model results

Model	RMSE	MAE	R ²	Bias
Bicubic interpolation	1.1220	0.3533	−0.286	0.009
XGBoost (no AEF)	0.9167	0.3710	0.141	−0.003
XGBoost + AEF	0.8957	0.3577	0.180	−0.009

Table 1: Single-step Pair A and Pair B downscaling (50 → 25 km and 25 → 12.5 km) on the 2025 test set, evaluated on reconstructed fine-resolution precipitation fields. XGBoost + AEF gives the best overall performance, improving RMSE and R² over both bicubic interpolation and XGBoost without AEF. This suggests that the AEF helps with the downscaling.

Metric	Mean	Std	Range
RMSE (ensemble mean)	4.7477	5.7561	[0.5515, 25.9738]
RMSE (bicubic baseline)	4.6820	5.9321	[0.3284, 26.2319]
CRPS	2.4044	4.0682	[0.1998, 17.5658]
Spread / Skill ratio	0.8855	0.5310	[0.0618, 1.6664]
Coverage (50%)	0.0429	0.0334	[0.0032, 0.1105]
Coverage (80%)	0.4156	0.1294	[0.0418, 0.5979]
Coverage (90%)	0.6495	0.1865	[0.0562, 0.8831]

Table 2: Recursive ensemble downscaling of the XGBoost + AEF baseline model (25 km $\rightarrow$ 1.5625 km, 4 steps) evaluated against PRISM 800 m observations over 30 random test days. Metrics are reported as mean, standard deviation, and minimum–maximum range across test days.

From the results in Table 2, the recursive XGBoost + AEF ensemble does not improve over bicubic interpolation in terms of RMSE: the ensemble mean RMSE is 4.7477, slightly higher than the bicubic baseline RMSE of 4.6820. This suggests that although the model performs reasonably in single-step downscaling, errors accumulate during recursive rollout and reduce its advantage at the final 1.5625 km scale. The CRPS value of 2.4044 also shows that the probabilistic forecast still has substantial error, especially on difficult precipitation days, as indicated by the large standard deviation and wide range.

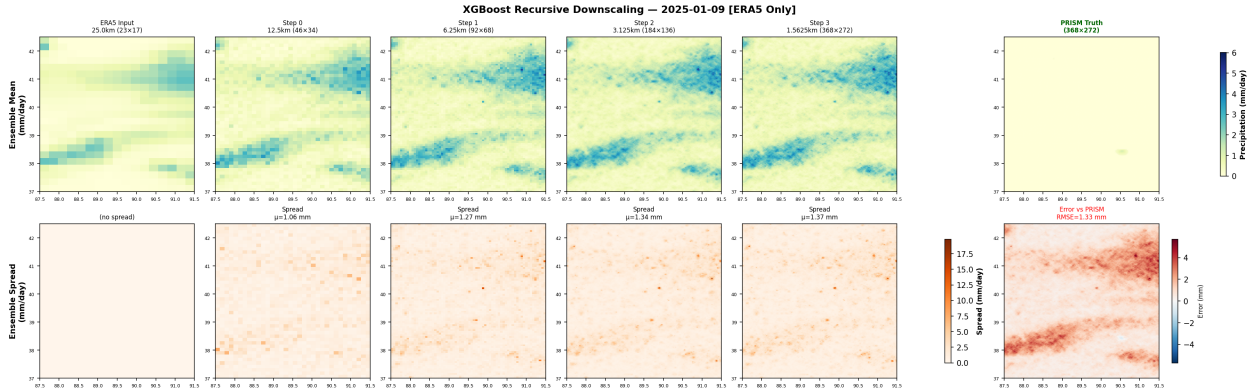


Figure 1: XGBoost recursive downscaling for Jan 9, 2025 (heavy precipitation only in ERA5). The recursive downscaling increases spatial detail from 25 km to 1.5625 km, with uncertainty gradually growing across steps from 1.06 mm to 1.37 mm, yet the final downscaled field differs substantially from PRISM truth with a large RMSE of 1.33 mm, indicating overprediction in XGBoost and bias between ERA5 and PRISM data.

5.2 CFM Model Results

During training, Phase 1’s flow loss converges while Phase 2 and Phase 3 have no meaningful loss convergence. Comparing the average RMSE on the test set, we get 4.3642 for Phase 1, 4.2741 for Phase 2, and 4.2740 for Phase 3, indicating a slight improvement in test accuracy through all phases of training. Additional training curves and per-day examples are provided in the Appendix.

Metric	Mean	Std	Range
RMSE (ensemble mean)	4.2740	6.4981	[0.1952, 26.9312]
RMSE (bicubic baseline)	4.5682	6.3477	[0.2366, 26.9950]
CRPS	2.3442	4.3986	[0.0536, 18.9843]
Spread / Skill ratio	0.5562	0.4721	[0.0102, 1.5060]
Coverage (50%)	0.2966	0.1518	[0.0001, 0.5662]
Coverage (80%)	0.5378	0.2170	[0.0020, 0.8305]
Coverage (90%)	0.5467	0.2169	[0.0023, 0.8423]

Table 3: Recursive ensemble downscaling of the CFM model (25 km $\rightarrow$ 1.5625 km, 4 steps) evaluated against PRISM 800m observations over 60 random test days. Compared with the bicubic baseline, the CFM ensemble achieves lower RMSE, with ensemble mean RMSE decreasing from 4.5682 to 4.2740. The CRPS is 2.3442, showing that the probabilistic forecast still has notable uncertainty error, especially given the large standard deviation and wide range across test days. The coverage values improve compared with the XGBoost baseline, especially at the 50% level, but they are still below the ideal 50%, 80%, and 90% coverage targets. This suggests that CFM gives better mean prediction than the baseline but still remains underdispersive and needs better uncertainty calibration.

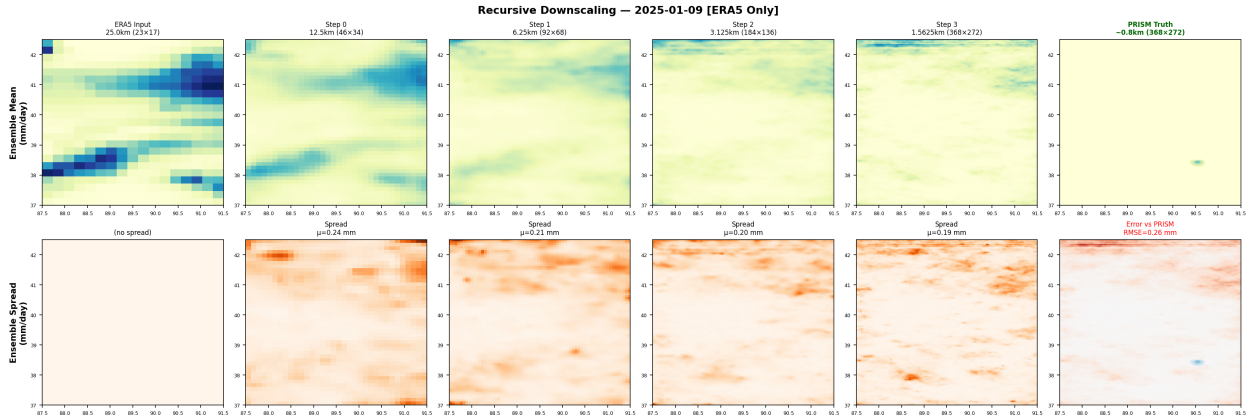


Figure 2: CFM recursive downscaling for Jan 9, 2025 (heavy precipitation only in ERA5). CFM produces a smoother and more conservative field that is much closer to PRISM, with a much lower RMSE of 0.26 mm vs. 1.33mm in XGBoost in this example. However, the ensemble spread decreases from 0.24 mm to 0.19 mm instead of increasing through the rollout, showing that CFM suppresses false precipitation but remains underdispersive and underestimates uncertainty.

5.3 CFM vs. Baselines

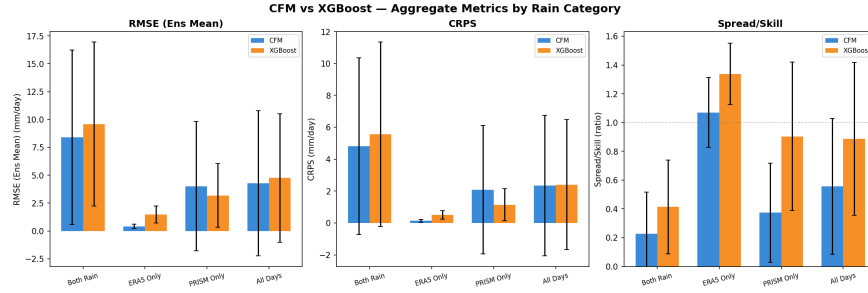


Figure 3: CFM vs. XGBoost aggregate metrics by rain category with RMSE, CRPS and Spread/Skill measures. CFM is generally more accurate than XGBoost, it tends to produce conservative, uncertain ensembles. XGBoost has better spread/skill outputs, but it tends to overestimate with high certainty inflating its ensemble validation metrics. CFM still requires better uncertainty calibration, which can be resolved with more training data.

6 Conclusions

This project built a recursive precipitation downscaling pipeline from 25 km ERA5 to 1.5625 km resolution over Illinois using AEF embeddings and PRISM validation. In single-step downscaling, XGBoost + AEF performs best, reaching $\text{RMSE} = 0.8957$ and $R^2 = 0.180$, showing that AEF provides useful spatial information. However, recursive downscaling is much harder. The XGBoost + AEF baseline has final $\text{RMSE} = 4.7477$, slightly worse than bicubic interpolation, and its ensemble coverage is far below the ideal levels. This shows that empirical residual noise injection cannot capture fine-scale precipitation uncertainty after multiple recursive steps. The CFM model improves recursive RMSE to about 4.274 and produces more realistic, conservative fields than XGBoost. Still, its uncertainty calibration remains limited, and Phase 2/3 training does not fully converge. Overall, our results show that AEF embeddings and CFM are promising for probabilistic climate downscaling, but larger training data and improved calibration are needed for stronger ensemble performance.

6.1 Limitations

- **Training Data:** Downloading and preprocessing data and embeddings was time intensive and with limited availability, guiding our decision to restrict terrain to the Illinois region. AEF embeddings are available only from 2017–2025, limiting training to 8 years.
- **Extension of Phase 3:** Phase 3 training resulted in no meaningful loss convergence. This phase could be extended to include more epochs and potential validation improvement against PRISM.

6.2 Future Work

- **Stochastic Model:** An inherently stochastic model such as diffusion to compare its CRPS and other ensemble coverage metrics with the current CFM model.
- **Different Foundation Models:** Expanding compatibility to other GeoFMs (Olmo, Prithvi-EO-2.0, Clay, TESSERA, etc.). This would also allow us to extend the data’s temporal range to pre-2017, currently limited by AEF.
- **Generalizable Scaling:** During downscaling steps, the CFM model is currently trained on 50 km $\rightarrow$ 25 km $\rightarrow$ 12.5 km... We would like to investigate smoothly downscaling to any resolution, as well as validating intermediate generation steps with other ground-truth datasets.

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7 Appendix

CFM Training Phases

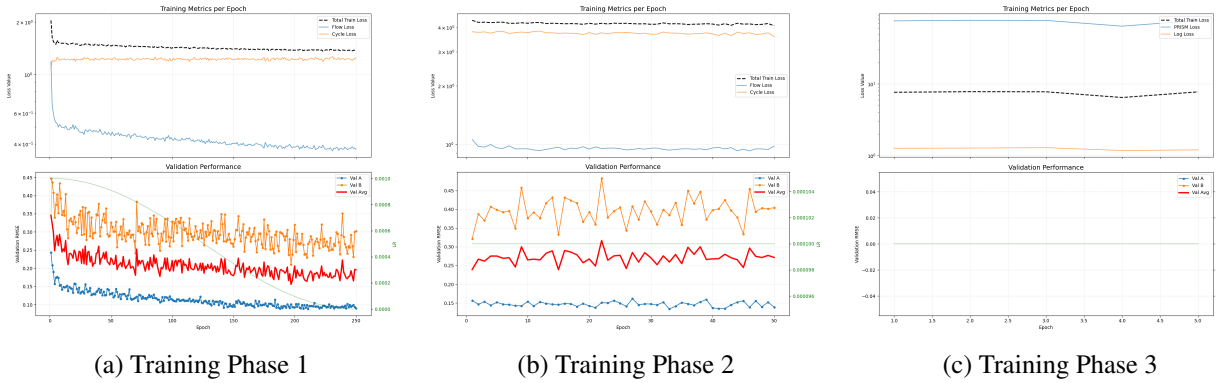


Figure 5: Comparison of the three CFM training phases. Training Phase 1’s flow loss converges, while Phase 2 and Phase 3 have minimal loss convergence. We also notice validation metrics remain noisy throughout training epochs, indicating potential overfitting to the training set or high model sensitivity. This can possibly be mitigated by adding more diverse training data through expanded temporal and spatial ranges.

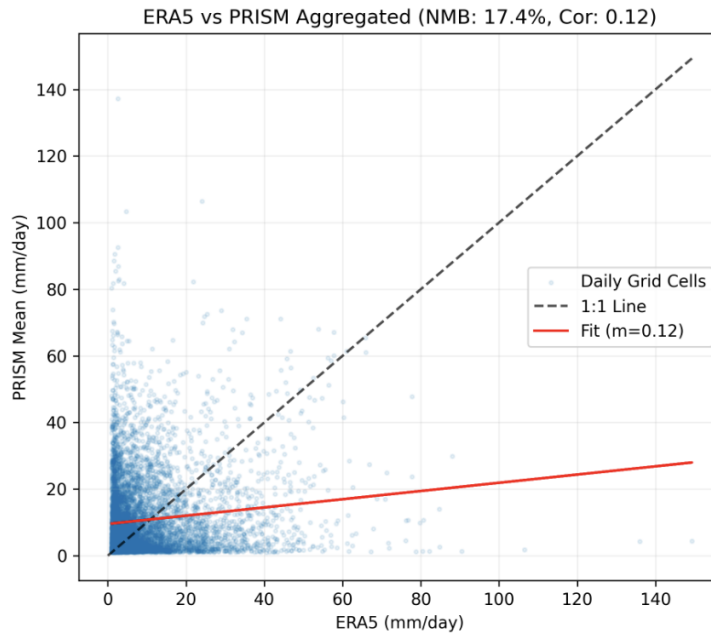


Figure 4: Pair C corrects the final 1.5625 km ERA5 output using PRISM as ground truth. The low ERA5–PRISM correlation indicates bias, spatial displacement, and smoothing of extreme precipitation, motivating an additional correction stage.

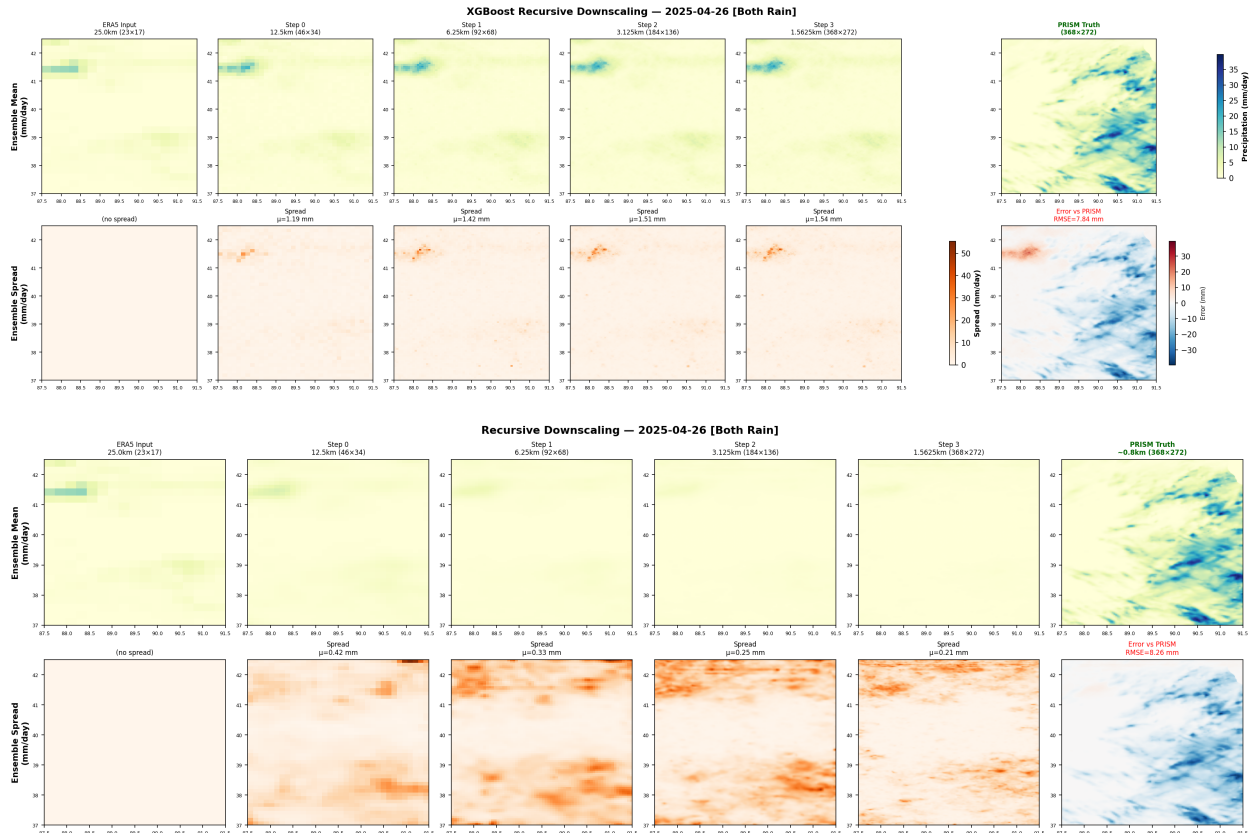


Figure 6: Example comparison of recursive downscaling between XGBoost and CFM model at April 26, 2025 (precipitation in both ERA5 and PRISM). The XGBoost baseline produces sharper but more biased spatial patterns, while the CFM model gives a smoother and more conservative prediction. These examples show that CFM still needs better uncertainty calibration.

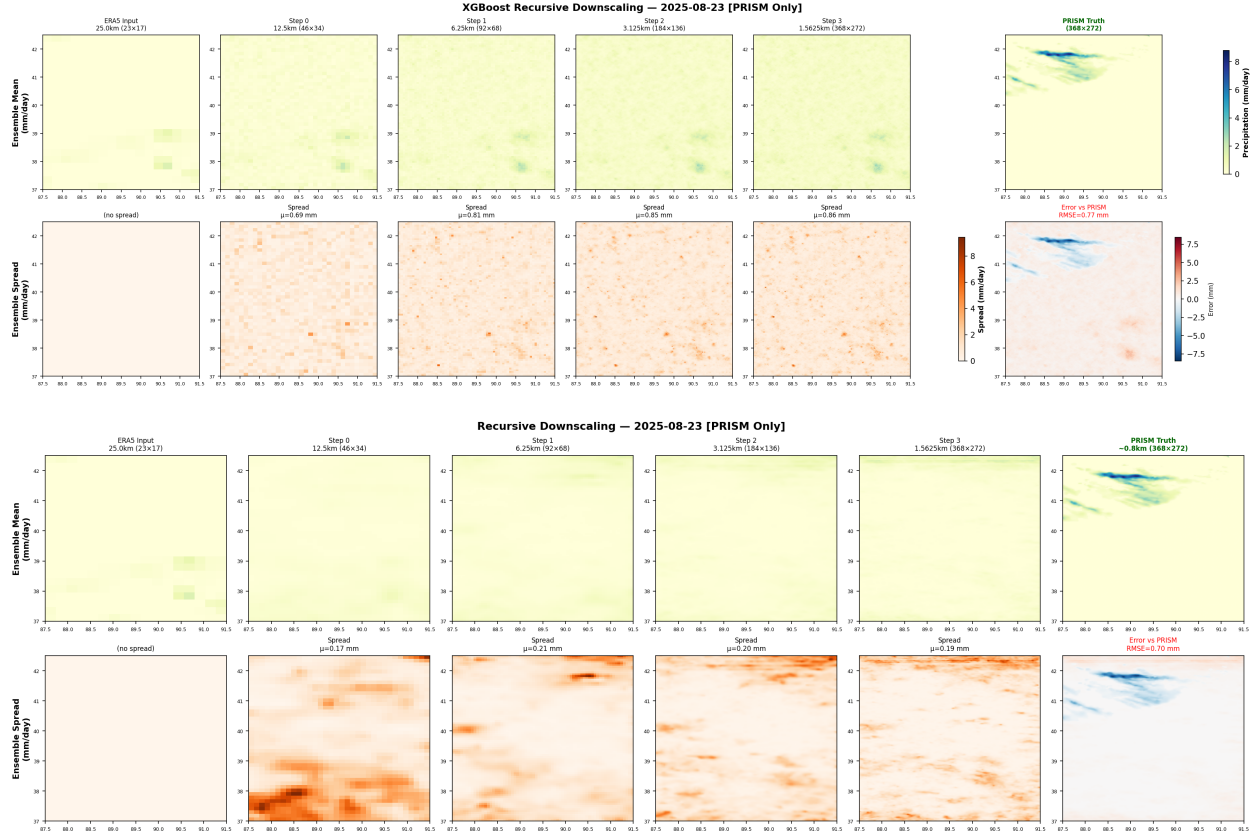


Figure 7: Example comparison of recursive downscaling between XGBoost and CFM model at Aug 23, 2025 (precipitation in only PRISM). The XGBoost adds sharper but misplaced rain patterns and CFM giving a smoother, more conservative prediction that still underestimates both precipitation and uncertainty.

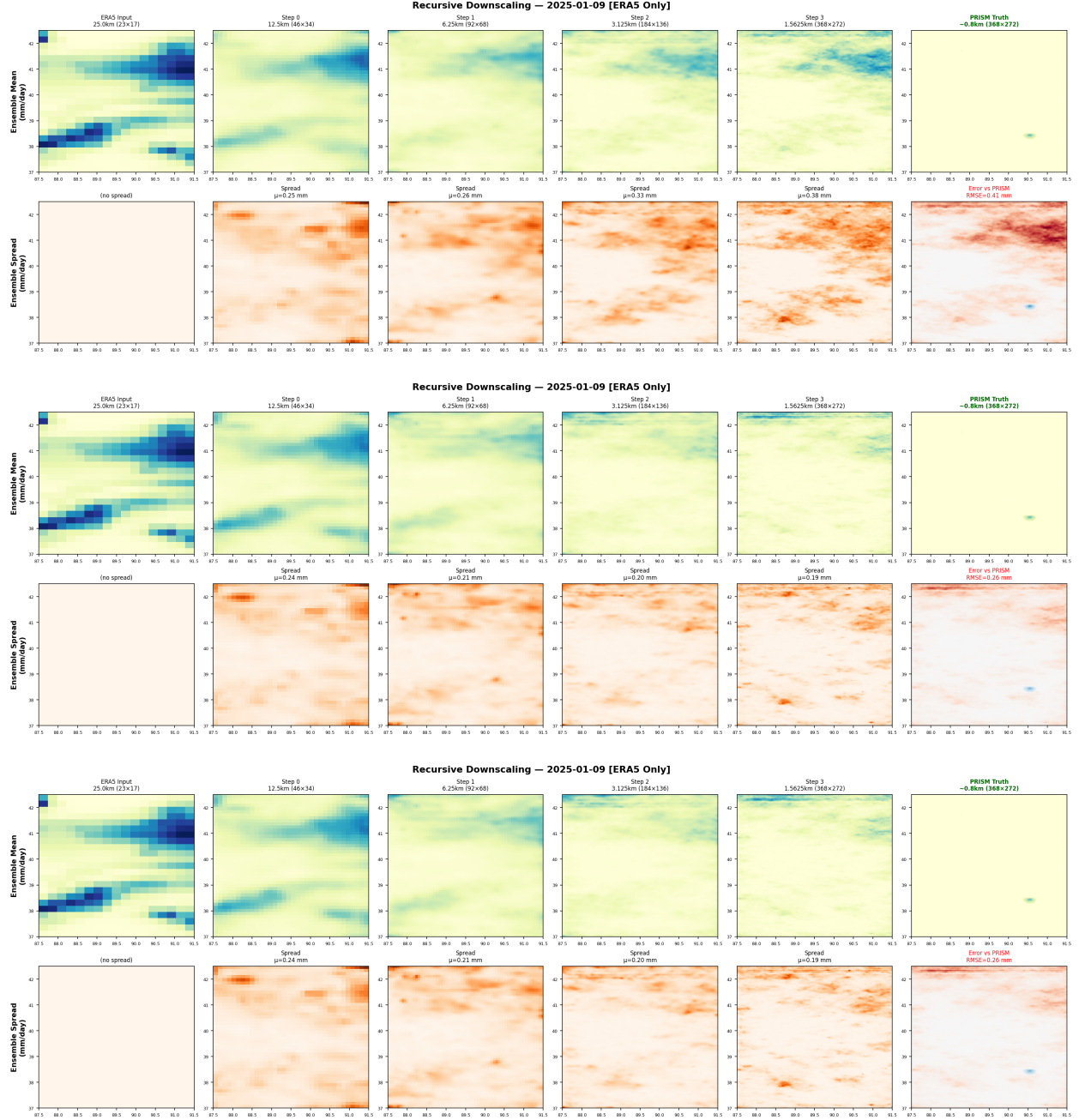


Figure 8: Example comparison of recursive downscaling between the 3 training phases in CFM model at Jan 9, 2025 (precipitation in only ERA5). Across the three CFM training phases, the model becomes slightly more conservative and closer to PRISM, especially from phase 1 to phase 2, and the almost same results for phase 2 and phase 3 indicates phase 3 did not learn very well. But still, all phases still retain ERA5-driven false precipitation patterns and underestimate uncertainty.

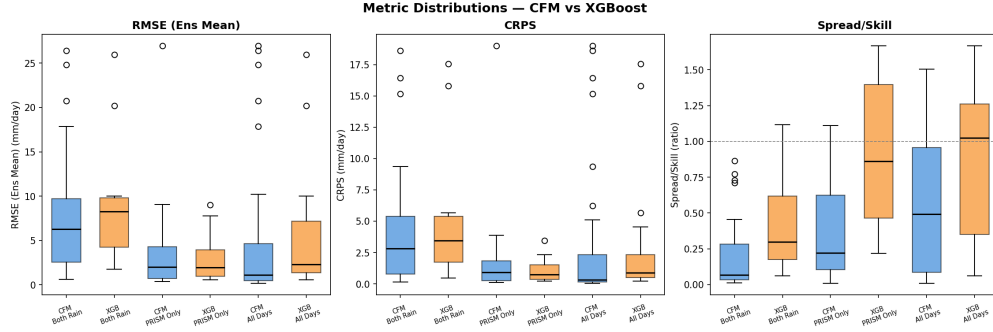


Figure 9: Distribution of RMSE, CRPS, and spread-skill ratio across evaluation days for CFM (blue) and XGBoost (orange), stratified by rain category. RMSE and CRPS medians are comparable between models for both_rain days, with CFM showing a slightly tighter interquartile range on PRISM-only days. The key distinction appears in spread-skill ratio (right panel, dashed line at 1.0 = perfect calibration): XGBoost achieves median spread-skill of 0.5–0.7 across categories, while CFM remains below 0.3, indicating that CFM ensembles are underdispersive — ensemble members cluster too tightly around the mean. This suggests the learned flow field captures the conditional mean well but underrepresents fine-scale precipitation variability, whereas XGBoost’s empirical noise injection better reflects the observed residual distribution.

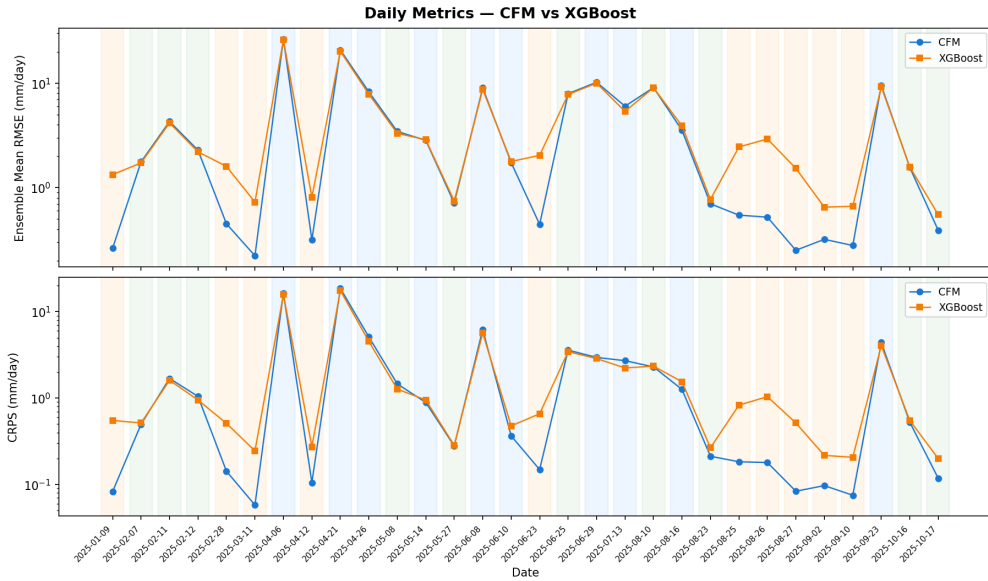


Figure 10: Per-day RMSE (top) and CRPS (bottom) for the CFM and XGBoost models across 30 matched evaluation days in 2025, shown on a log scale. Background shading indicates rain category: blue = both_rain (ERA5 and PRISM agree on rain), orange = era5_only, green = prism_only. Both models track closely on RMSE, with the largest errors (>10 mm/day) occurring on heavy convective days where ERA5 underestimates precipitation (e.g., April 6, April 21). CFM achieves lower RMSE on light-rain and dry days but comparable or slightly higher errors on extreme events. CRPS, which penalizes both bias and ensemble miscalibration, shows a similar pattern, with CFM’s narrower ensembles occasionally yielding worse probabilistic scores despite high point-estimate accuracy.